

SAS Project Submission

Category-2

Q1) Explain about ETL Process and define it. Provide a Detailing on phases of ETL?

Ans1) ETL in SAS studio means Extract, Transformation and Load. It describes about extracting the data which is first step of the process, Transforming the data, filtering with no errors, joining etc. And loading which is the final step of getting a cleaned and prepared dataset. ETL is a integration process used in SAS to combine different datasets from different sources into a single dataset with no errors and storing it into a SAS library for later use.

 **Extract:** It's the first step in the process, in this step user joins different datasets from different sources. User can use functions such as 'Proc Import' to extract datasets. 'Libname' statement to create new library and store all datasets into it. 'Proc sql' to read, edit, manipulate the data. User can join tables using point and click functions or use code window to perform various joints such as left join, full merge/join etc.

 **Transformation:** In this step cleaning and validating of data is done. Using functions such as recode, describe missing values, changing data types, perform calculations etc.

 **Load:** Final step is to load the clean, validated data into databases or libraries. Data can be stored into temporary or permanent libraries based on the requirement. Statements like Proc Sql, Libname, Proc Export are used in this step.

Q2) Explain the process of Exploratory Data Analysis (EDA)?

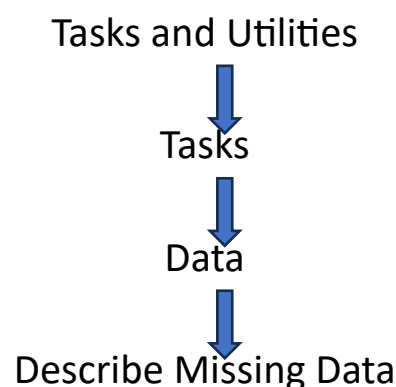
Ans2) Exploratory Data Analysis (EDA) in SAS is the process to examine a dataset to understand its structure, detect any anomalies, identify patterns before building models or making a project. It makes easier for user to understand and present data in a orderly manner.

- 'Proc contents' statement in SAS helps to identify variable names, types, and their attributes.
- 'Proc means' and 'Proc freq' statements gives us mean, median, mode, standard deviation, variance, range, min and max value etc. Point and click functions under data tab in task and utilities window also gives summary statistics for a dataset.
- Graphical representation of data can be done to present data in a more user friendly and understandable way. Different graphs like histogram, box plot, pie chart, scatter plot, bar and line chart etc.
- Describe and detect missing values, duplicate values, filtering and sorting of data is also done using various tools under data tab in task window.
- Assumptions can be made using linear regressions, binary logistic regressions, covariance etc under linear models tab in tasks window. This to check relation and variance between variables in a dataset.

Q3) Explain the task about Describe missing data? With flow chart representation? Create an example?

Ans3) This function in SAS helps user to identify and describe missing data in a dataset. It is a critical part in data cleaning and preprocessing. Can be done using statements or point and click functions in SAS studio.

 **Point and Click:** This is a easy and convenient way.



SAS® Studio

*sasbpnew *Describe Missing Data

Settings Code/Results Split

DATA INFORMATION

DATA

WORK.IMPORT

Filter: (none)

ROLES

*Analysis variables: (50 items)

- MedCtrlID
- Status
- Death Cause
- Sex
- BP Status
- Weight
- cholesterol level

ADDITIONAL ROLES

CODE LOG RESULTS

```
1 * Task code generated by SAS Studio 3.8
2 *
3 * Generated on '13/05/2025 23:02'
4 * Generated by 'u64183310'
5 * Generated on server 'ODAWS02-APSE1-2.ODA.SAS.COM'
6 * Generated on SAS platform 'Linux X64 5.14.0-284.30.1.el9_2.x86_64'
7 * Generated on SAS version '9.04.01M7P88062020'
8 * Generated on browser 'Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/125.0.0.0 Safari/537.36'
9 * Generated on web client 'https://odamid-apse1-2.oda.sas.com/SASStudio/main?locale=en_GB&zone=...'
10 */
11
12
13
14
15 ods noproctitle;
16
17 proc format;
18   value _nmissprint low-high="Non-missing";
19   value $_cmissprint " "="" other="Non-missing";
20 run;
21
22 proc freq data=WORK.IMPORT;
23   title3 "Missing Data Frequencies";
24   title4 h=2 "Legend: ., A, B, etc = Missing";
25   format MedCtrlID Weight 'cholesterol level' Diastolic Systolic _nmissprint.;
26   format Status 'Death Cause' Sex 'BP Status' $_cmissprint.;
27   tables MedCtrlID Status 'Death Cause' Sex 'BP Status' Weight
28         'cholesterol level' Diastolic Systolic / missing nocum;
29 run;
30
```

Messages: 2 User: u64183310 UTF-8

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- cholesterol level

ADDITIONAL ROLES

CODE LOG RESULTS

Errors, Warnings, Notes

Errors

Warnings

Notes (13)

```
118 proc delete data=Work._MissingData_;
119 run;
```

NOTE: Deleting WORK._MISSINGDATA_ (memtype=DATA).
NOTE: PROCEDURE DELETE used (Total process time):

real time	0.00 seconds
user cpu time	0.00 seconds
system cpu time	0.00 seconds
memory	184.87k
OS Memory	21668.00k
Timestamp	13/05/2025 05:32:39 PM
Step Count	35
Page Faults	0
Page Reclaims	15
Page Swaps	0
Voluntary Context Switches	15
Involuntary Context Switches	0
Block Input Operations	0
Block Output Operations	0

```
120
121 OPTIONS NONOTES NOSTIMER NOSOURCE NOSYNTAXCHECK;
```

Messages: 2 User: u64183310

Missing Data Frequencies

Legend: ., A, B, etc = Missing

MedCtrlID		
MedCtrlID	Frequency	Percent
Non-missing	5	100.00

Status		
Status	Frequency	Percent
	1	20.00
Non-missing	4	80.00

Death Cause		
Death Cause	Frequency	Percent
	1	20.00
Non-missing	4	80.00

Sex		
Sex	Frequency	Percent
	1	20.00
Non-missing	4	80.00

BP Status		
BP Status	Frequency	Percent
	1	20.00
Non-missing	4	80.00

Weight		
Weight	Frequency	Percent
.	1	20.00
Non-missing	4	80.00

cholesterol level		
cholesterol level	Frequency	Percent
Non-missing	5	100.00

Diastolic		
Diastolic	Frequency	Percent
.	2	40.00
Non-missing	3	60.00

Systolic		
Systolic	Frequency	Percent
Non-missing	5	100.00

Missing Data Patterns across Variables

Legend: ., A, B, etc = Missing

<https://odamid-apse1-2.oda.sas.com/SASStudio/sasexec/submissions/6194ef88-f641-4f46-9086-28c5101577f9/results>

13/05/2025, 23:02

Results: Describe Missing Data

MedCtrlID	Status	Death Cause	Sex	BP Status	Weight	cholesterol level	Diastolic	Systolic	Frequency	Percent
Non-missing		Non-missing	Non-missing	Non-missing	Non-missing	Non-missing	.	Non-missing	1	20
Non-missing	Non-missing		Non-missing	Non-missing	Non-missing	Non-missing	.	Non-missing	1	20
Non-missing	Non-missing	Non-missing		Non-missing	.	Non-missing	Non-missing	Non-missing	1	20
Non-missing	Non-missing	Non-missing	Non-missing		Non-missing	Non-missing	Non-missing	Non-missing	1	20
Non-missing	Non-missing	Non-missing	Non-missing	Non-missing	Non-missing	Non-missing	Non-missing	Non-missing	1	20

This is a example on Describing Missing Data. It shows frequency (number of missing values) and its percentage in the table.

Q4) Explain the task about Recode or Standardize values for gender specification? Create an Example?

Ans4) Recode task in SAS changes or transforms a variable into another variable. Can be applied to both characters and numeric variables. It can also convert text values to numeric codes, or replacing specific values. Easy to use **Task and Utilities > Tasks > Data > Recode values.**

SAS® Studio

Server Files and Folders

Tasks and Utilities

My Tasks

Tasks

Data

List Table Attributes

Characterize Data

Describe Missing Data

List Data

Transpose Data

Stack/Split Columns

Filter Data

Select Random Sample

Partition Data

Sort Data

Rank Data

Transform Data

Standardize Data

Recode Values

Recode Ranges

Combine Tables

Snippets

Libraries

File Shortcuts

*sasbpnew * * * * Recode Values

DATA VALUES INFORMATION

DATA

WORK.IMPORT

ROLES

Recode:

☐ Numeric variable

☒ Character variable

Variable to recode: (1 item)

Sex

OUTPUT DATA SET

Recoded variable name:

Recoded Gender

☐ Write to input data set

☒ Write to another data set

Data set name:

work.recodedValues0001

Browse

CODE LOG RESULTS OUTPUT DATA

Log Code

```

1 /*
2 *
3 * Task code generated by SAS Studio 3.8
4 *
5 * Generated on '13/05/2025 23:16'
6 * Generated by 'u64183310'
7 * Generated on server 'ODAWS02-APSE1-2.ODA.SAS.COM'
8 * Generated on SAS platform 'Linux LIN X64 5.14.0-284.30.1.el9_
9 * Generated on SAS version '9.04.01M7P08062020'
10 * Generated on browser 'Mozilla/5.0 (Windows NT 10.0; Win64; x6
11 * Generated on web client 'https://odamid-apse1-2.oda.sas.com/S
12 *
13 */
14
15 data work.recodedValues0001;
16   length 'Recoded Gender'n $ 6;
17   set WORK.IMPORT;
18
19   select (Sex);
20     when ('M') 'Recoded Gender'n='Male';
21     when ('F') 'Recoded Gender'n='Female';
22     otherwise 'Recoded Gender'n=Sex;
23   end;
24 run;

```

UTF-8

Messages: 2 User: u64183310

SAS® Studio

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Partition Data

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Rank Data

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Recode Ranges

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File Shortcuts

*sasbpnew * * * * Recode Values

DATA VALUES INFORMATION

VALUES

Recode values: (minimum 1 row)

Old value	New value
M	Male
F	Female

CODE LOG RESULTS OUTPUT DATA

Log Code

Errors, Warnings, Notes

Errors

Warnings

Notes (3)

```

92 run;

```

NOTE: There were 5 observations read from the data set WORK.IMPORT

NOTE: The data set WORK.RECODEDVALUES0001 has 5 observations and 1

NOTE: DATA statement used (Total process time):

	real time	user cpu time	system cpu time	memory	OS Memory	Timestamp	Step Count	Page Faults	Page Reclaims	Page Swaps	Voluntary Context Switches	Involuntary Context Switches	Block Input Operations	Block Output Operations
	0.00 seconds	0.00 seconds	0.01 seconds	956.12k	21420.00k	13/05/2025 05:46:59 PM	31	0	169	0	12	0	0	264

93

94

OPTIONS NONOTES NOSTIMER NOSOURCE NOSYNTAXCHECK;

Messages: 2 User: u64183310

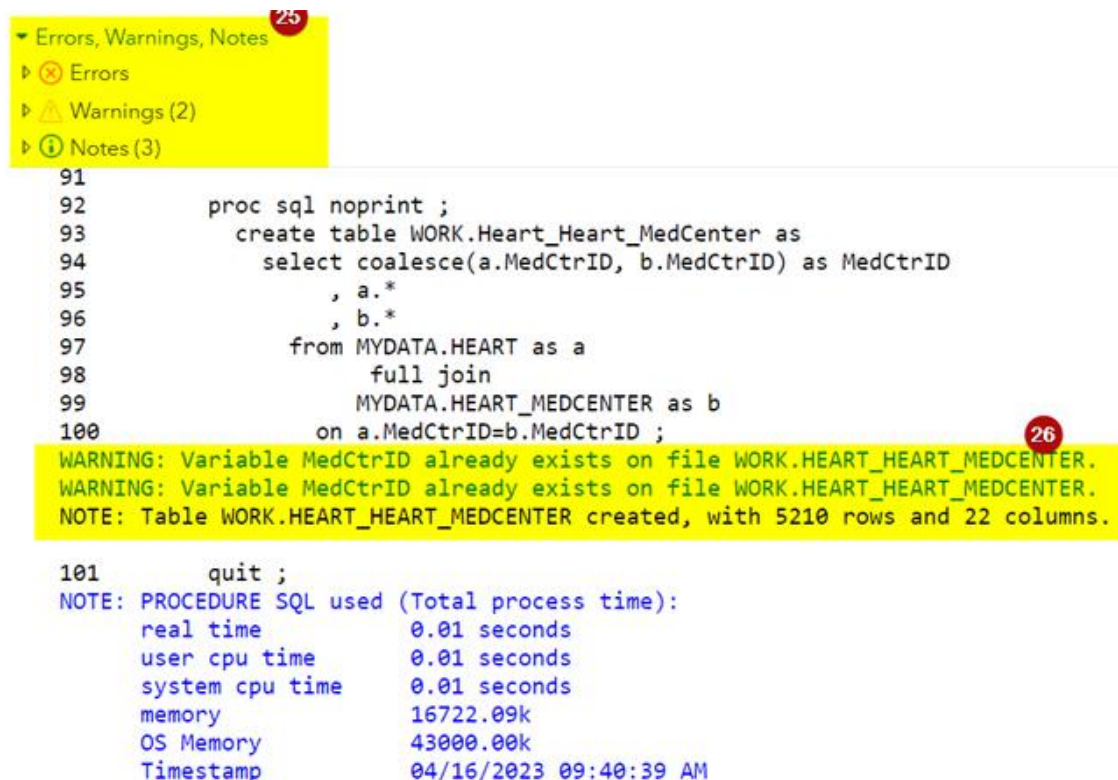
13/05/2025, 23:17

Results: WORK.RECODEDVALUES0001

Obs	Recoded Gender	MedCtrlID	Status	Death Cause	Sex	BP Status	Weight	cholesterol level	Diastolic	Systolic
1	Male	1	alive	None	M	120/80	72	200	73	122
2	Male	2	alive	None	M	120/85	79	220	75	120
3	Female	3	dead	low BP	F	90/60	69	190	72	119
4	Female	4	alive	None	F	40/90	65	230	79	121
5	Male	5	alive	None	M	120/80	73	235	81	120

This is the example on Gender Specification using recode function. Here 'M' and 'F' under column 'sex' is converted into 'Male' and 'Female' and stored under column Recoded Gender.

Q5) Explain the below mentioned sample with output results?



```
25
▼ Errors, Warnings, Notes
  ▶ ✖ Errors
  ▶ ⚠ Warnings (2)
  ▶ ⓘ Notes (3)

91
92      proc sql noprint ;
93          create table WORK.Heart_Heart_MedCenter as
94              select coalesce(a.MedCtrID, b.MedCtrID) as MedCtrID
95                  , a.*
96                  , b.*
97              from MYDATA.HEART as a
98                  full join
99                  MYDATA.HEART_MEDCENTER as b
100             on a.MedCtrID=b.MedCtrID ;
26
WARNING: Variable MedCtrID already exists on file WORK.HEART_HEART_MEDCENTER.
WARNING: Variable MedCtrID already exists on file WORK.HEART_HEART_MEDCENTER.
NOTE: Table WORK.HEART_HEART_MEDCENTER created, with 5210 rows and 22 columns.

101      quit ;
NOTE: PROCEDURE SQL used (Total process time):
      real time           0.01 seconds
      user cpu time       0.01 seconds
      system cpu time     0.01 seconds
      memory              16722.09k
      OS Memory           43000.00k
      Timestamp           04/16/2023 09:40:39 AM
```

Ans5) Above mentioned program is to combine two tables. And merge performed is full join. It joins all the columns from table A to table B. Above example is done using point and click function: **Tasks and Utilities > Tasks > Data > Combine Tables**. This function allows user to combine two datasets into one dataset.

SAS® Studio

*sas_reg *sas-reg-2 *Combine Tables Program 1

CODE LOG RESULTS OUTPUT DATA

```
1 /*
2 *
3 * Task code generated by SAS Studio 3.8
4 *
5 * Generated on '13/05/2025 23:50'
6 * Generated by 'u64183310'
7 * Generated on server 'ODAWS02-APSE1-2.ODA.SAS.COM'
8 * Generated on SAS platform 'Linux LIN X64 5.14.0-284.30.1.el9_2.x86_64'
9 * Generated on SAS version '9.04.01M7P08062020'
10 * Generated on browser 'Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/136.0.0.0 Safari/537.36'
11 * Generated on web client 'https://odamid-apse1-2.oda.sas.com/SASStudio/main?locale=en_GB&zone=GHT%252B05%253A30&ticket=ST-109096-GTAB3o6o1gLXLrghSuTv-cas'
12 *
13 */
14
15 /* The DATA step and PROC SQL methods produce the same results when the match column */
16 /* values uniquely identify each row and the tables have no other columns with the same name. */
17 /* Otherwise, the results may differ. */
18
19 proc sql noprint;
20 create table work.heart as
21 select coalesce(a.'Med Cnt Id'n, b.'Med Cnt Id'n) as MedCntId
22 , a.*
23 , b.*
24 from work.import as a
25 full join
26 work.import1 as b
27 on a.'Med Cnt Id'n=b.'Med Cnt Id'n;
28 quit;
```

Line 21, Column 42 UTF-8

Messages: 15 User: u64183310

SAS® Studio

Server Files and Folders

Tasks and Utilities

- My Tasks
- Tasks
 - Data
 - List Table Attributes
 - Characterize Data
 - Describe Missing Data
 - List Data
 - Transpose Data
 - Stack/Split Columns
 - Filter Data
 - Select Random Sample
 - Partition Data
 - Sort Data
 - Rank Data
 - Transform Data
 - Standardize Data
 - Recode Values
 - Recode Ranges
 - Combine Tables
- Snippets
- Libraries
- File Shortcuts

*sas_reg *sas-reg-2 *Combine Tables Program 1

CODE LOG RESULTS OUTPUT DATA

Errors, Warnings, Notes

Errors

Warnings (13)

Notes (3)

```
87
88 proc sql noprint;
89 create table work.heart as
90 select coalesce(a.'Med Cnt Id'n, b.'Med Cnt Id'n) as MedCntId
91 , a.*
92 , b.*
93 from work.import as a
94 full join
95 work.import1 as b
96 on a.'Med Cnt Id'n=b.'Med Cnt Id'n;
WARNING: Variable 'Med Cnt ID'n already exists on file WORK.HEART.
WARNING: Variable 'Age CHD Diag'n already exists on file WORK.HEART.
WARNING: Variable 'Age At Start'n already exists on file WORK.HEART.
WARNING: Variable Height already exists on file WORK.HEART.
WARNING: Variable Weight already exists on file WORK.HEART.
WARNING: Variable Diastolic already exists on file WORK.HEART.
WARNING: Variable Systolic already exists on file WORK.HEART.
WARNING: Variable MRW already exists on file WORK.HEART.
WARNING: Variable 'Smoking Status'n already exists on file WORK.HEART.
WARNING: Variable 'Age At Death'n already exists on file WORK.HEART.
WARNING: Variable Cholesterol already exists on file WORK.HEART.
WARNING: Variable Sex already exists on file WORK.HEART.
WARNING: Variable 'Death Cause'n already exists on file WORK.HEART.
NOTE: Table WORK.HEART created, with 5 rows and 14 columns.
```

Messages: 15 User: u64183310

14/05/2025, 00:07

Results: WORK.HEART

Obs	MedCntId	Med Cnt ID	Age CHD Diag	Age At Start	Height	Weight	Diastolic	Systolic	MRW	Smoking Status	Age At Death	Cholesterol	Sex	Death Cause
1	1	1	45	39	152	75	75	121	32.5	Heavy Smoker	51	190	Male	Chronic HD
2	2	2	36	32	165	68	72	119	25	Light Smoker	0	225	Female	None
3	3	3	41	36	160	73	80	120	28.5	Moderate Smoker	45	200	Female	Chronic HD
4	4	4	32	30	165	76	76	115	27.9	Non Smoker	0	230	Male	None
5	5	5	30	29	163	70	80	120	27	Non Smoker	0	221	Male	None

—THE END—